

Supply Chain Visibility System with Optimization Analytics

1. Student Details

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Project Title: Supply Chain Visibility System with Optimization Analytics

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2. Introduction

This project is a Supply Chain Visibility System with Optimization Analytics, built in Power BI on top of a real-world AIS (Automatic Identification System) vessel-tracking dataset enriched with hourly weather data.

The project is needed because organizations moving cargo across a network often lack a single, unified view of fleet position, transit speed, delays, and cost — information that today sits scattered across raw tracking feeds. The dashboard solves this by turning raw vessel-movement pings into KPIs and visuals that surface fleet utilization, delivery reliability, inventory-style throughput, and supplier/transportation cost patterns at a glance.

Across Milestone 1, 2, 3, and 4, the project implements: a cleaned star-schema data model with foundational KPIs (Milestone 1); an inventory- and delivery-style implementation dashboard (Milestone 2); a supplier and transportation cost/quality analytics page (Milestone 3); and a warehouse analytics view with a final consolidated dashboard (Milestone 4).

The project aims to provide end-to-end visibility into inventory, suppliers, warehouses, transportation and delivery operations using Power BI.

3. Tools Used

- Power BI
- Power Query
- DAX
- Microsoft Excel
- Data Visualization and Data Modelling

4. Milestone 1 – Data Modelling & KPI Foundation

4.1 Milestone 1 Introduction

This milestone establishes the analytical foundation for the Supply Chain Visibility System: importing and cleaning the source AIS + weather dataset, designing a star-schema data model, defining core KPIs with DAX, and building an initial dashboard that gives visibility into fleet movement and cargo flow through the supply chain network.

4.2 Dataset Overview

The project uses an AIS (Automatic Identification System) vessel-tracking dataset enriched with hourly weather data. The dataset contains 1,098,967 records covering approximately 10,754 unique vessels tracked on 31 March 2022, with fields including vessel position (latitude/longitude), speed over ground, course, estimated time of arrival, destination cluster, and weather conditions (temperature, wind speed, precipitation, visibility) at each ping.

Note on scope: this dataset tracks cargo vessels in transit rather than warehouse stock (no SKU, order, or COGS fields exist). Core KPIs in Section 5 are therefore adapted from classic inventory-management metrics to their maritime supply-chain equivalents — most notably, Cargo Throughput Velocity is used as the direct analogue of inventory turnover, representing how fast cargo moves through the network rather than how fast stock cycles through a warehouse.

4.2.1 Dataset Schema

Full column list as loaded into Power BI (41 columns, 1,098,967 rows):

Column(s)	Data Type	Description
MMSI	Whole Number	Maritime Mobile Service Identity — unique vessel ID
BaseDateTime	Date/Time	Timestamp of the AIS ping
LAT / LON	Decimal Number	Vessel latitude / longitude at time of ping
SOG / SOG_kmh	Decimal Number	Speed over ground (knots / km per hour)
COG / Heading	Decimal Number	Course over ground / compass heading
VesselName / IMO / CallSign	Text	Vessel identifiers

Column(s)	Data Type	Description
VesselType / VesselType_enc	Text / Whole Number	Vessel category, label and encoded form
Status / Status_enc	Text / Whole Number	AIS navigation status, label and encoded form
Length / Width / Draft	Decimal Number	Vessel physical dimensions
Cargo / Cargo_enc	Text / Whole Number	Cargo type, label and encoded form
TransceiverClass	Text	AIS transceiver class (A/B)
dest_cluster / dest_lat / dest_lon	Whole Number / Decimal	Destination hub cluster ID and coordinates
dist_km	Decimal Number	Distance to destination in km
ETA_min / ETA_hours	Decimal Number	Estimated time of arrival
Speed_Category	Text	Derived label: Fast / Moderate / Slow / Stopped
timestamp_str, lat_r, lon_r, group_id	Text / Decimal	Supporting fields from source pipeline
weather_available	Whole Number	Flag indicating a weather record was matched
weather_temp_c / weather_humidity	Decimal Number	Temperature and humidity at ping location/hour
weather_precip_mm / weather_wind_kph	Decimal Number	Precipitation and wind speed
weather_wind_dir_degree / weather_pressure_mb	Decimal Number	Wind direction and atmospheric pressure
weather_cloud / weather_visibility_km	Decimal Number	Cloud cover and visibility

4.3 Data Import & Cleaning

Data was imported into Power BI via Get Data → Text/CSV and cleaned in Power Query. Key cleaning actions:

Field / Issue	Action Taken	Reasoning
VesselName (~0.2% blank)	Replaced with "Unknown Vessel"	Ping still carries valid GPS/speed data
IMO (~9% blank)	Replaced with "N/A"	Small vessels legitimately lack an IMO number

Field / Issue	Action Taken	Reasoning
CallSign (~1.7% blank)	Replaced with "N/A"	Non-critical identifier field
weather_visibility_km (~3% blank)	Left null / excluded from weather visuals	Avoids fabricating values
Duplicate pings	Removed on MMSI + BaseDateTime	Same ship, same timestamp = transmission duplicate
SOG_kmh outliers	Filtered to < 100 km/h	Removes unrealistic AIS speed spikes

4.3.1 Derived Columns

Additional derived columns were created for downstream KPI logic:

- **TripDate** — date component extracted from BaseDateTime
- **HourOfDay** — hour component extracted from BaseDateTime
- **IsMoving** — "Moving" / "Stationary" flag based on SOG_kmh > 1.85 km/h (~1 knot)
- **DelayFlag** — binary flag for trips with above-threshold ETA_hours (ETA_hours > 40)

4.3.2 Power Query (M) — Fact and Dimension Queries

The single cleaned query was split into a fact query and four referenced dimension queries.

Simplified M code for each:

Fact_VesselMovement (base cleaning, abbreviated):

```
let
    Source = Csv.Document(File.Contents("ais_data_complete_weather.csv")),
    ChangedType = Table.TransformColumnTypes(Source, {
        {"BaseDateTime", type datetime}, {"SOG_kmh", type number}, ...}),
    RemovedDup = Table.Distinct(ChangedType, {"MMSI", "BaseDateTime"}),
    FilteredSpeed = Table.SelectRows(RemovedDup, each [SOG_kmh] < 100),
    AddedTripDate = Table.AddColumn(FilteredReaderSpeed, "TripDate",
        each Date.From([BaseDateTime])),
    AddedHour = Table.AddColumn(AddedTripDate, "HourOfDay",
        each Time.Hour([BaseDateTime])),
    AddedIsMoving = Table.AddColumn(AddedHour, "IsMoving",
        each if [SOG_kmh] > 1.85 then "Moving" else "Stationary"),
    AddedDelayFlag = Table.AddColumn(AddedIsMoving, "DelayFlag",
        each if [ETA_hours] > 40 then 1 else 0)
in
    AddedDelayFlag
```

Dim_Vessel (referenced from Fact_VesselMovement):

```
let
  Source = Fact_VesselMovement,
  KeptColumns = Table.SelectColumns(Source, {"MMSI", "VesselName", "IMO",
    "CallSign", "VesselType", "Length", "Width", "Draft", "Cargo", "TransceiverClass"}),
  RemovedDup = Table.Distinct(KeptColumns, {"MMSI"})
in
  RemovedDup
```

Dim_Destination (referenced from Fact_VesselMovement):

```
let
  Source = Fact_VesselMovement,
  KeptColumns = Table.SelectColumns(Source, {"dest_cluster", "dest_lat", "dest_lon"}),
  RemovedDup = Table.Distinct(KeptColumns, {"dest_cluster"})
in
  RemovedDup
```

Dim_Status (referenced from Fact_VesselMovement):

```
let
  Source = Fact_VesselMovement,
  KeptColumns = Table.SelectColumns(Source, {"Status", "Status_enc"}),
  RemovedDup = Table.Distinct(KeptColumns, {"Status_enc"}),
  AddedLabel = Table.AddColumn(RemovedDup, "Status_Label", each
    if [Status_enc] = 0 then "Under way using engine"
    else if [Status_enc] = 1 then "At anchor"
    else if [Status_enc] = 5 then "Moored"
    else "Other / Undefined")
in
  AddedLabel
```

4.4 Supply Chain Data Model

The cleaned flat table was restructured into a star schema: a single fact table capturing vessel movement events, connected to four dimension tables via one-to-many relationships (single-direction filtering). This structure mirrors a standard supply-chain / logistics data model, adapted to a vessel-tracking context.

4.4.1 Model Structure

- **Fact_VesselMovement** — the transactional table: one row per AIS ping (position, speed, ETA, weather at time of ping)
- **Dim_Vessel** — one row per vessel (MMSI, name, type, dimensions, cargo type)
- **Dim_Destination** — one row per destination cluster (cluster ID, lat/lon)
- **Dim_Status** — one row per navigation status code (status label, encoded value)
- **Dim_Date** — a standalone calendar table (Date, Year, MonthName, Weekday, DayNum)

Dim_Date was built directly in Model view using DAX rather than referenced from the fact query:

```
Dim_Date = CALENDAR(DATE(2022,3,1), DATE(2022,4,30))
Year = YEAR(Dim_Date[Date])
MonthName = FORMAT(Dim_Date[Date], "MMMM")
Weekday = FORMAT(Dim_Date[Date], "dddd")
DayNum = DAY(Dim_Date[Date])
```

4.4.2 Relationship Diagram

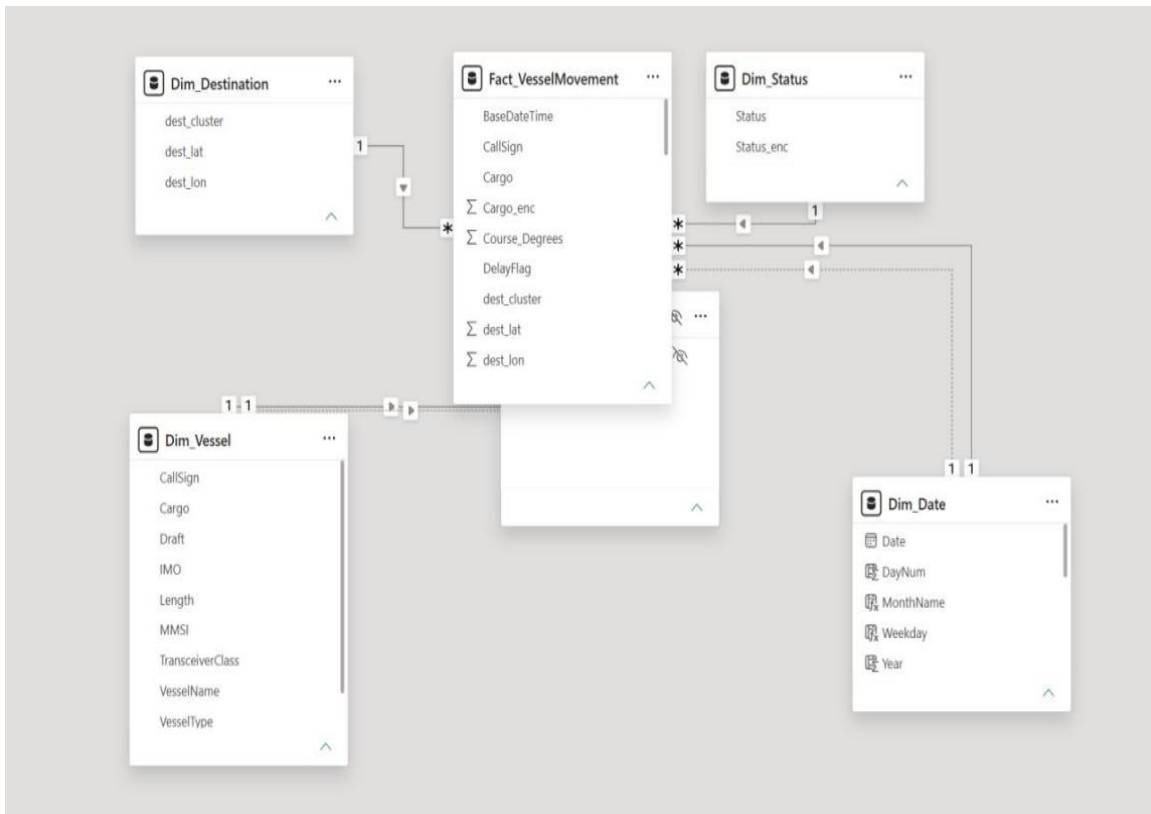


Figure 1: Star schema — Fact_VesselMovement connected to Dim_Vessel, Dim_Destination, Dim_Status, and Dim_Date

4.5 Core KPIs & DAX Measures

All measures are stored in a dedicated _Measures table, kept separate from the fact and dimension tables for clarity.

Measure	DAX Formula	Business Meaning
Total Vessels	DISTINCTCOUNT(Dim_Vessel[MMSI])	Unique ships tracked in the network

Measure	DAX Formula	Business Meaning
Total Pings	COUNTROWS(Fact_VesselMovement)	Raw event volume / data-quality check
Avg Speed (km/h)	AVERAGE(Fact_VesselMovement[SOG_kmh])	Average vessel speed over ground
Avg Transit Distance (km)	AVERAGE(Fact_VesselMovement[dist_km])	Average distance covered per trip
Avg ETA (hours)	AVERAGE(Fact_VesselMovement[ETA_hours])	Average estimated time of arrival
% Stationary Vessels	DIVIDE(CALCULATE(COUNTROWS(...), IsMoving="Stationary"), [Total Pings])	Share of fleet idle at a given time
Fleet Utilization Rate	DIVIDE(CALCULATE(DISTINCTCOUNT(MMSI), IsMoving="Moving"), [Total Vessels])	Share of fleet actively in transit
Cargo Throughput Velocity	DIVIDE([Avg Transit Distance], [Avg ETA (hours)])	Inventory-turnover equivalent: speed of cargo flow through the network
Weather-Impacted Trips %	DIVIDE(CALCULATE(COUNTROWS(...), precip_mm>0), [Total Pings])	Share of trips occurring under precipitation
Avg Wind Speed (kph)	AVERAGE(Fact_VesselMovement[weather_wind_kph])	Average wind conditions across tracked trips

Measure	DAX Formula	Business Meaning
Delayed Trips %	<code>DIVIDE(CALCULATE(COUNTROWS(...), DelayFlag=1), [Total Pings])</code>	Proxy for on-time delivery

4.6 Inventory Turnover Dashboard

The dashboard below — titled "Cargo Turnover & Fleet Visibility Dashboard" — brings together the KPI cards, geographic vessel map, bottleneck analysis, fleet utilization breakdown, and weather-impact view.

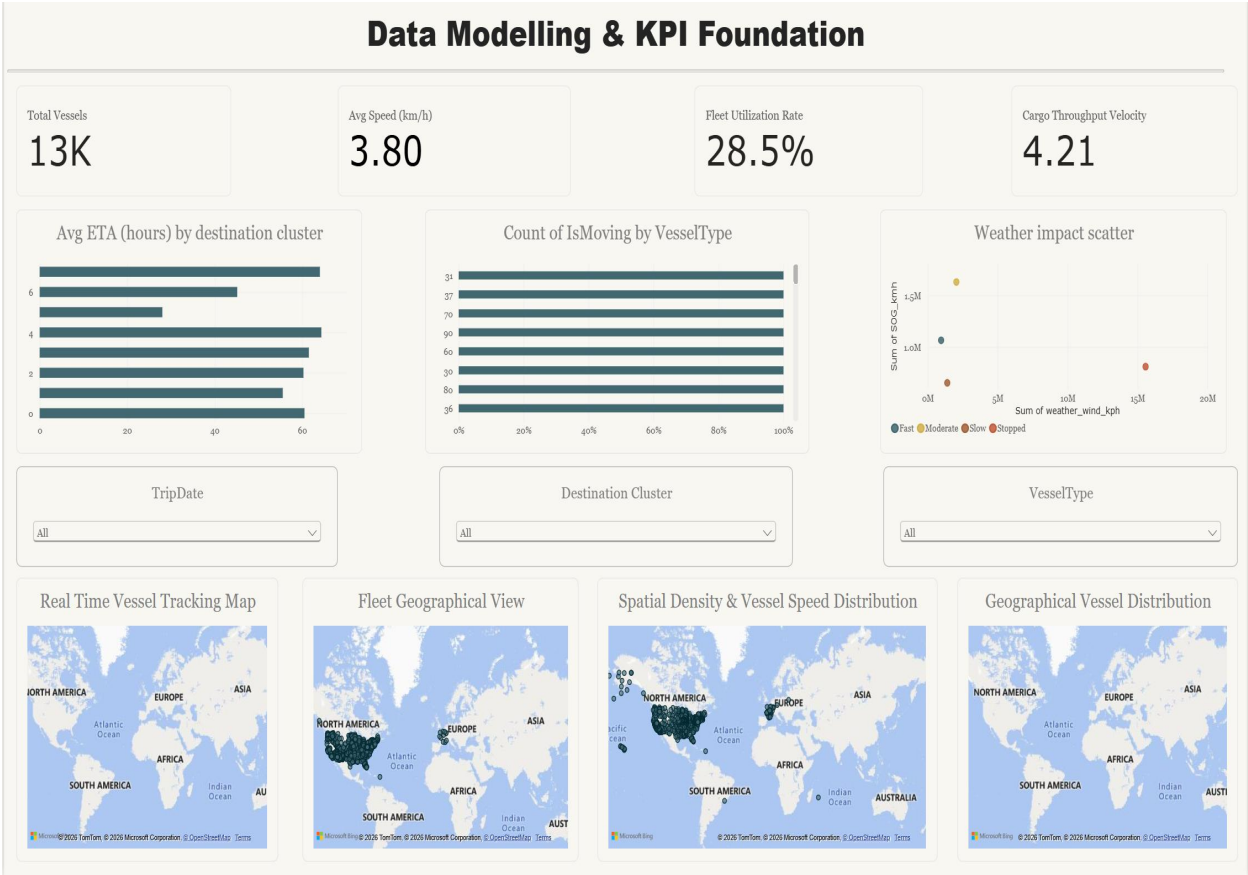


Figure 2: Data Modelling & KPI Foundation — Power BI

4.6.1 Dashboard Components

- KPI cards: Total Vessels (13K), Avg Speed (3.80 km/h), Fleet Utilization Rate (28.5%), Cargo Throughput Velocity (4.21)
- Avg ETA (hours) by destination cluster — bar chart highlighting slower-transit hubs

- Count of IsMoving by VesselType — moving vs. stationary breakdown across vessel categories
- Weather impact scatter — Sum of weather_wind_kph vs. Sum of SOG_kmh, colored by Speed_Category (Fast / Moderate / Slow / Stopped)
- Geographic map visuals — vessel positions by latitude/longitude and speed category
- Slicers — TripDate, dest_cluster, VesselType for interactive filtering

Explanation of dashboard

Milestone 1 establishes the analytical foundation of the Supply Chain Visibility System by transforming a large AIS vessel-tracking dataset enriched with weather information into a structured Power BI data model. The dataset contains approximately **1,098,967 records and 10,754 unique vessels**, with information such as vessel location, speed, destination, ETA, navigation status, cargo type, and weather conditions. The main insight from this milestone is that **data quality, modelling structure, and KPI definition are critical for obtaining reliable supply-chain visibility**.

The cleaning process addressed several important data-quality issues. Duplicate AIS pings were removed using the combination of MMSI and timestamp, unrealistic vessel speeds were filtered, and missing identifier fields were handled without fabricating operational values. Additional fields such as **TripDate, HourOfDay, IsMoving, and DelayFlag** were created to support later analysis. This creates a consistent foundation for distinguishing active vessel movement from stationary conditions and identifying potential delays.

A major insight is the importance of the **star-schema model**. Instead of keeping all information in a single flat table, the project separates vessel movement events from vessel, destination, status, and date dimensions. This structure allows the dashboard to analyze the same movement data from multiple operational perspectives while maintaining consistent relationships. The KPI results provide an initial picture of network performance.

The dashboard reports an average vessel speed of approximately **3.80 km/h**, a **Fleet Utilization Rate of 28.5%**, and a **Cargo Throughput Velocity of 4.21**. In this project, Cargo Throughput Velocity acts as an inventory-turnover equivalent because the dataset does not contain conventional warehouse inventory or SKU information.

The most important operational insight is the relatively low fleet utilization rate. Approximately **28.5% of the distinct vessels are classified as actively moving**, meaning that a large proportion of vessels are stationary at the measured point.

However, this should not automatically be interpreted as operational inefficiency because the dataset includes port-adjacent AIS activity where anchored or moored vessels are expected. Therefore, the result primarily establishes a **baseline for future optimization rather than proving that the stationary vessels represent avoidable idle time**.

Overall, Milestone 1 demonstrates that the project has established a reliable analytical foundation. The cleaned dataset, star schema, DAX measures, and baseline KPIs make it possible to move from raw vessel-tracking information toward inventory, delivery, transportation, and optimization analysis in the subsequent milestones.

5. Milestone 2 – Inventory & Delivery Analytics

5.1 Milestone 2 Introduction

This milestone builds the Weeks 3–4 Implementation Output page on top of the Milestone 1 data model, reusing the existing fact/dimension tables and _Measures table, with new measures and one new calculated column (DeliveryStatus). Since the reference layout is a retail inventory/delivery dashboard (Product, Category, Warehouse, Inventory Value, Turnover Ratio) and the dataset has no literal product, warehouse stock, or shipment record, the following mapping is used:

The reference layout used for this milestone is a classic retail inventory/delivery dashboard (Product, Category, Warehouse, Inventory Value, Turnover Ratio). The project dataset is AIS vessel-tracking and weather data, which has no literal product, warehouse stock, or shipment record. To keep the same visual layout and KPI names as the brief while staying accurate about the underlying data, the following mapping is used throughout this milestone — the same approach applied to Cargo Throughput Velocity in Milestone 1:

Reference Dashboard Term	Dataset Equivalent
Product	Individual vessel (VesselName / MMSI)
Category	VesselType
Warehouse	dest_cluster (destination hub)
Region	dest_cluster grouped, or dest_lat / dest_lon
Inventory Value	Proxy: total distance in transit (SUM(dist_km)) — represents "asset flow volume" instead of stock value
Inventory Turnover	Cargo Throughput Velocity (from Milestone 1)
Shipments	AIS pings / trips (COUNTROWS(Fact_VesselMovement))
On-time / Delayed Delivery	DelayFlag (already built in Milestone 1)
Delivery Time (Days)	ETA_hours

5.2 Inventory Analysis

5.2.1 New DAX Measures

The following measures were added to the _Measures table for this milestone:

```
Total Inventory = SUM(Fact_VesselMovement[dist_km])

Total Products = DISTINCTCOUNT(Dim_Vessel[MMSI])

Slow-Moving Products =
CALCULATE(
    DISTINCTCOUNT(Fact_VesselMovement[MMSI]),
    Fact_VesselMovement[Speed_Category] = "Slow"
)
```

Cargo Throughput Velocity is reused directly from Milestone 1 as the Inventory Turnover KPI for this dashboard.

5.2.2 KPI Cards (Top Row)

Four card visuals summarize the network at a glance:

- **Total Inventory** — 277.09M (total distance-in-transit proxy for flow volume)
- **Inventory Turnover** — 4.21 (Cargo Throughput Velocity)
- **Total Products** — 13K (unique vessels tracked)
- **Slow-Moving Products** — 3K (vessels in the "Slow" speed category)

5.2.3 Bar Chart — Category vs Inventory

A clustered bar chart shows Total Inventory by VesselType ("Total Inventory by VesselType"), sorted descending, identifying which vessel categories account for the largest share of network flow volume.

5.2.4 Column Chart — Warehouse vs Inventory

A clustered column chart ("Inventory by Warehouse") plots Total Inventory against dest_cluster, highlighting that one destination hub carries a disproportionately large share of tracked flow compared to the others.

5.2.5 Line Chart — Inventory Trend

A line chart tracks Total Inventory by Date ("Total Inventory by Date"), giving a time-based view of flow volume across the tracked period.

5.3 Delivery Analysis

5.3.1 Donut Chart — On-Time vs Delayed Deliveries

A new calculated column converts the DelayFlag into a readable label:

```
DeliveryStatus = IF(Fact_VesselMovement[DelayFlag] = 1, "Delayed", "On-time")
```

The donut chart ("On-Time vs Delayed Deliveries") shows a total of 1,095,540 pings, split into 213.69K on-time (19.51%) and 881.85K delayed (80.49%), giving a proxy for on-time delivery performance across the network. This on-time share of roughly 20% is the same figure that reappears as the On-Time Rate KPI in Milestone 3.

5.3.2 Bar Chart — Region vs Delivery Performance

A clustered bar chart ("Delivery Performance by Region") plots the Delayed Trips % measure against dest_cluster (used as "Region"), surfacing which destination hubs experience the highest share of delayed trips.

5.3.3 Line Chart — Delivery Trend

A line chart ("On-time Delivery Trend") tracks delivery performance over time, showing the trend rising through the middle of the tracked window before tapering off toward the end.

5.4 Drill-Down

5.4.1 Hierarchy

A three-level Location Hierarchy was built on dest_cluster (Dim_Destination), with VesselType and then VesselName added beneath it:

- Level 1 — Warehouse (dest_cluster)
- Level 2 — Category (VesselType)
- Level 3 — Product (VesselName)

5.4.2 Application

The Location Hierarchy was applied to the axis of a clustered column chart ("Inventory by Warehouse (Drill to Category & Product)") with Total Inventory as the values field. Using the drill-down controls, the chart re-renders from destination-hub totals down to individual vessel

type and then individual vessel level, confirming the hierarchy functions correctly across all three levels.

5.5 Trend and Variance Analysis

5.5.1 New DAX Measures

Because the dataset spans a short real-world window, variance is computed hour-over-hour rather than week-over-week:

```
Prev Hour Throughput =
CALCULATE(
    [Cargo Throughput Velocity],
    FILTER(ALL(Fact_VesselMovement[HourOfDay]),
        Fact_VesselMovement[HourOfDay] = MAX(Fact_VesselMovement[HourOfDay]) - 1)
)

Throughput Variance = [Cargo Throughput Velocity] - [Prev Hour Throughput]
Throughput Variance % = DIVIDE([Throughput Variance], [Prev Hour Throughput])
```

5.5.2 Line Chart — Turnover Trend Analysis

A dual-line chart ("Turnover Trend Analysis") plots Inventory Turnover alongside Throughput Variance % by hour, showing turnover recovering and climbing through the later hours of the tracked window after an early dip.

5.5.3 Table — Product | Category | Warehouse | Inventory | Turnover

A detail table lists vessels (Product) with Category, Warehouse, Total Inventory, and Inventory Turnover, sorted so the entries can be scanned by turnover rate. The table totals 27,70,89,265.68 in Total Inventory and 4.21 in overall Inventory Turnover across all rows.

Product	Category	Warehouse	Total Inventory	Inventory Turnover
RUMAWAY	36	6	689.91	0.26
DAGNY	0	6	890.94	0.33
NOMAD	40	3	3,177.04	0.36
BROOKE CHAPMAN	31	5	1,559.12	0.39
BLAKE	36	6	52.84	0.41
ATLS 9701	90	3	2,536.13	0.42
MARY CATHERINE	31	5	4,359.12	0.43

5.6 Milestone 2 Dashboard

The completed Weeks 3–4 Implementation Output page, combining the Inventory Analysis, Delivery Analysis, Drill-Down, and Trend & Variance Analysis sections described above, is shown below:



Figure 3: Milestone 2 — Inventory & Delivery Analytics Dashboard (Power BI)

Explanation of dashboard

Milestone 2 extends the foundation created in Milestone 1 by translating the AIS vessel-tracking dataset into an inventory and delivery analytics framework. Because the dataset does not contain traditional products, warehouses, inventory values, or shipment records, the project uses clearly defined operational equivalents. Individual vessels represent products, vessel types represent categories, destination clusters represent warehouses, total distance in transit represents an inventory-value proxy, and Cargo Throughput Velocity represents inventory turnover.

The dashboard reports Total Inventory of 277.09M, representing the total distance-in-transit proxy, while Inventory Turnover is 4.21. The dashboard also identifies approximately 13K unique vessels and around 3K slow-moving vessels. These values indicate that the network contains a substantial amount of vessel movement, but a meaningful portion of vessels fall into the slow-moving category. This creates an opportunity for further investigation into the destinations, vessel categories, and operating conditions associated with slower movement. One of the clearest findings is the concentration of flow across destination hubs. The Inventory by Warehouse visualization shows that one destination cluster carries a disproportionately large share of tracked flow. This is important because concentration in a particular destination can make that hub particularly significant for overall network performance. If delays or congestion occur in a highly concentrated destination, the effect can potentially influence a large portion of the observed network flow.

The delivery analysis provides another significant insight. The On-Time vs Delayed Deliveries visualization contains approximately 1.095 million pings, with about 213.69K classified as on-time (19.51%) and 881.85K classified as delayed (80.49%). The On-Time Delivery Trend shows performance increasing through the middle portion of the tracked period before declining toward the end. Similarly, the Turnover Trend Analysis indicates that turnover recovers and increases during the later hours after an initial decline. These patterns demonstrate the usefulness of time-based monitoring because overall averages alone cannot show when operational performance changes. The drill-down functionality provides another important analytical benefit. Users can move from destination cluster → vessel type → individual vessel, allowing high-level flow patterns to be investigated at a more granular level. This means that an unusual inventory or delivery pattern identified at the destination level can be traced down to specific vessel categories.

6. Milestone 3 – Supplier & Transportation Analytics

This milestone extends the Milestone 1 data model and Milestone 2 measures into a Transportation & Supplier Analytics page. It reuses the same Fact_VesselMovement / Dim_Vessel / Dim_Destination tables and the _Measures table, and adds no new columns or tables — every new figure is a measure or an existing field reused.

6.1 Mapping the Brief to the Dataset

The Milestone 3 brief (Supplier, Carrier, Route, Cost, Quality Score) has no direct counterpart in the AIS + weather dataset. As in Milestones 1 and 2, each brief term is mapped to a declared proxy, and every proxy is labeled as such:

Brief term	Proxy from the data	Assumption
Carrier	VesselName	The vessel itself acts as the carrier
Supplier	Cargo (cargo type)	No supplier entity exists — cargo type is the closest "who is shipping what" grouping
Route	dest_cluster	Each destination hub represents a distinct route endpoint
Region	dest_cluster	Same field as Route — no separate geographic field was created
Transportation Cost	dist_km × assumed rate	Assumed \$2.50 per km — a modeling assumption, not real cost data
Shipment	One vessel–destination combination	No order/shipment ID exists; counted with SUMMARIZE, no column added
Fulfillment Rate	1 – % Stationary Vessels	Proxy: share of pings where vessels are actively moving rather than stalled
On-Time Rate	1 – Delayed Trips %	Real, not a proxy — computed directly from DelayFlag
Quality Score	100 × (1 – Delayed Trips %)	Proxy: reliability (low delay rate) stands in for quality

6.2 Dashboard

The completed Supplier and Transportation Analytics page, with four KPI cards, four slicers and four charts (each a different visual type), is shown below:

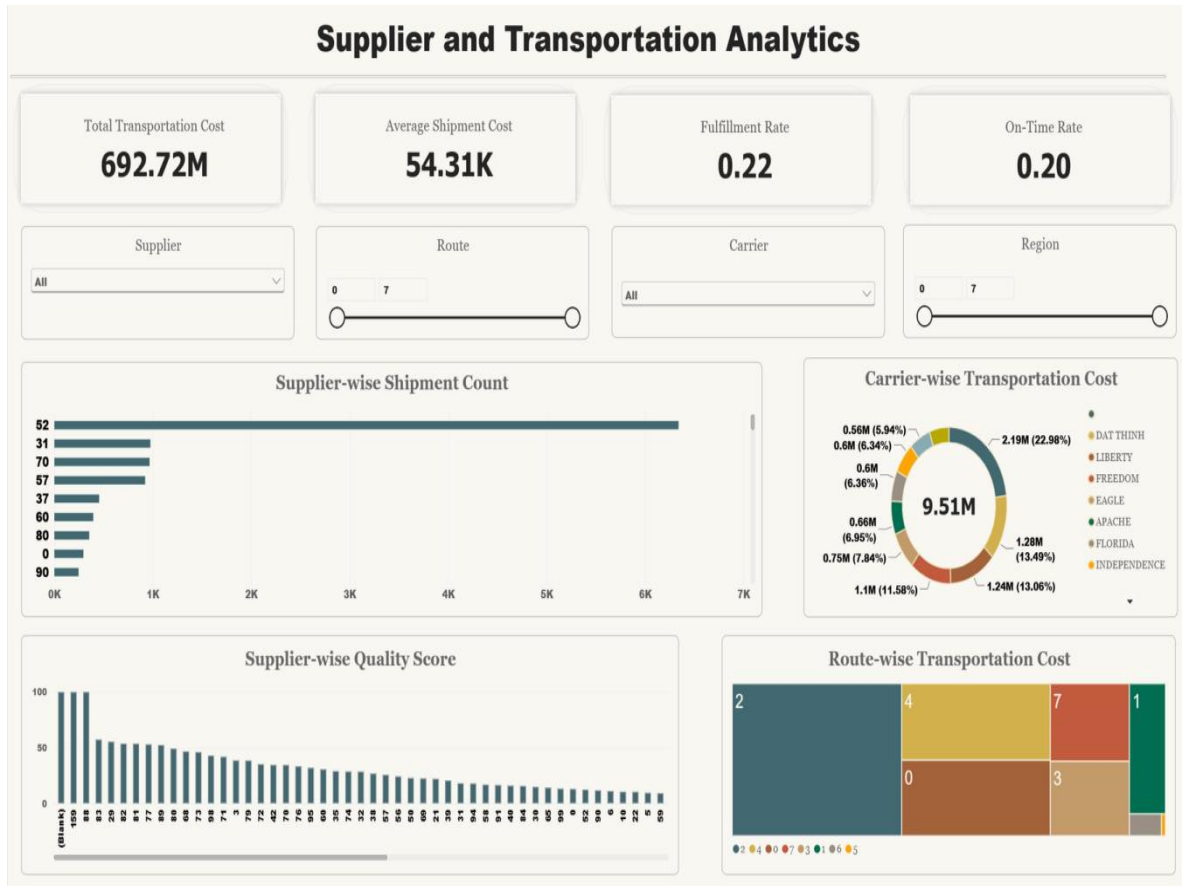


Figure 4: Milestone 3 — Supplier and Transportation Analytics Dashboard (Power BI)

6.3 Explanation

The following measures were added to the _Measures table. Those marked ASSUMPTION rely on a declared proxy or assumed value:

```
-- ASSUMPTION: $2.50 per km, for demonstration only, not real freight pricing
Assumed Cost Rate = 2.5

Total Transportation Cost =
SUMX(Fact_VesselMovement, Fact_VesselMovement[dist_km] * [Assumed Cost Rate])

Total Shipments =
COUNTROWS(
    SUMMARIZE(Fact_VesselMovement,
        Fact_VesselMovement[MMSI], Fact_VesselMovement[dest_cluster])
)

Average Shipment Cost = DIVIDE([Total Transportation Cost], [Total Shipments])

-- REAL, not a proxy: reuses the Milestone 1 DelayFlag
On-Time Rate = 1 - [Delayed Trips %]

-- ASSUMPTION: fulfillment = share of pings actively moving, not stalled
Fulfillment Rate = 1 - [% Stationary Vessels]

-- ASSUMPTION: quality = inverse of delay rate
Quality Score = 100 * (1 - [Delayed Trips %])
```

6.4 KPI Cards

KPI	Value	What it shows
Total Transportation Cost	692.72M	Modeled cost of 277.09M km of distance at the assumed \$2.50/km. The 277.09M km matches Total Inventory in Milestone 2, confirming the two pages reconcile.
Average Shipment Cost	54.31K	Total cost divided by roughly 12.8K vessel–destination combinations (the shipment proxy).
Fulfillment Rate	0.22 (22%)	Proxy. About 22% of pings show a vessel actively moving; the remaining ~78% are stationary.
On-Time Rate	0.20 (20%)	Real. About 20% of pings are on time, matching the 19.51% on-time share in the Milestone 2 donut.

Fulfillment Rate (a share of pings) and Fleet Utilization Rate from Milestone 1 (28.5%, a share of distinct vessels) are related but computed on different bases, so they are close without being equal.

6.5 Charts

Supplier-wise Shipment Count (clustered bar). Shipments are counted per Cargo type. The Cargo field is stored as numeric AIS codes, so suppliers appear as code values. Category 52 dominates with roughly 6.3K shipments — about six times the next categories (31, 70 and 57, each close to 1K). Shipment volume is therefore heavily concentrated in a single supplier category.

Supplier-wise Quality Score (clustered column). Quality Score is $100 \times$ the on-time share, so higher means fewer delayed pings. Most categories sit between roughly 10 and 57, well below the ideal of 100, consistent with the network-wide 20% on-time rate. Three categories ((Blank), 159 and 88) score 100 because none of their pings were delayed; these are most likely small-volume categories, so the score should be read with their low counts in mind.

Carrier-wise Transportation Cost (donut, Top 10). A Top N filter limits the donut to the ten highest-cost carriers, which together total 9.51M — only about 1.4% of the 692.72M network total. Within that group the largest slice — its legend label appears blank in the source data, so it may combine several unnamed vessels — accounts for 22.98% (2.19M), followed by DAT THINH (13.49%, 1.28M), LIBERTY (13.06%, 1.24M) and FREEDOM (11.58%, 1.10M). The percentages are shares of the top-10 total, not of the whole network. The small top-10 share shows that modeled cost is spread across thousands of vessels rather than dominated by a few.

Route-wise Transportation Cost (treemap). Destination cluster 2 is by far the largest block, followed by clusters 4 and 0, then 7, 3 and 1, with clusters 6 and 5 barely visible. Cluster 2 alone represents roughly two-fifths of modeled cost, mirroring the concentration seen in the Milestone 2 "Inventory by Warehouse" chart. Because cost is distance $\times$ a constant rate, the route ranking is the distance ranking scaled by 2.5, not an independent finding.

6.5 Slicers

Four slicers allow the whole page to be filtered: **Supplier** (Cargo, dropdown), **Carrier** (VesselName, dropdown), **Route** (dest_cluster, range 0–7) and **Region** (dest_cluster, range 0–7). Route and Region are bound to the same underlying field, so they are two labeled entry points to one filter rather than independent dimensions. Each slicer cross-filters all four KPI cards and all four charts.

7. Milestone 4 – Warehouse Analytics & Final Dashboard

Milestone 4 centers on broadening the current supply chain analytics model into covering warehouse efficiency, executive overview, and performance improvement. The previous AIS-based model for shipment data from earlier milestones was used to create warehouse-related KPIs and interactive Power BI dashboards.

The milestone has three dashboards:

1. **Warehouse Efficiency Report**
2. **Executive Overview**
3. **Performance Optimization**

The supplied AIS dataset does not contain specific warehouse management fields like order ID, warehouse capacity, physically shipped quantity, and picking transactions; therefore, the relevant analytical proxies were devised based on shipment distance, vessel movement, delay status, and ETA.

7.1 Data Model and Star Schema

Milestone 4 uses the present star-schema model. The main fact table is represented by Fact_Shipment which includes Data_Mart as:

- **Dim_Warehouse** – information about a destination/warehouse
- **Dim_Product** – information about the vessel and cargo
- **Dim_Status** – information about the status of the vessel/shipment
- **Dim_Date** – date-based analysis
- **_Measures** – DAX measures used in the KPI calculations

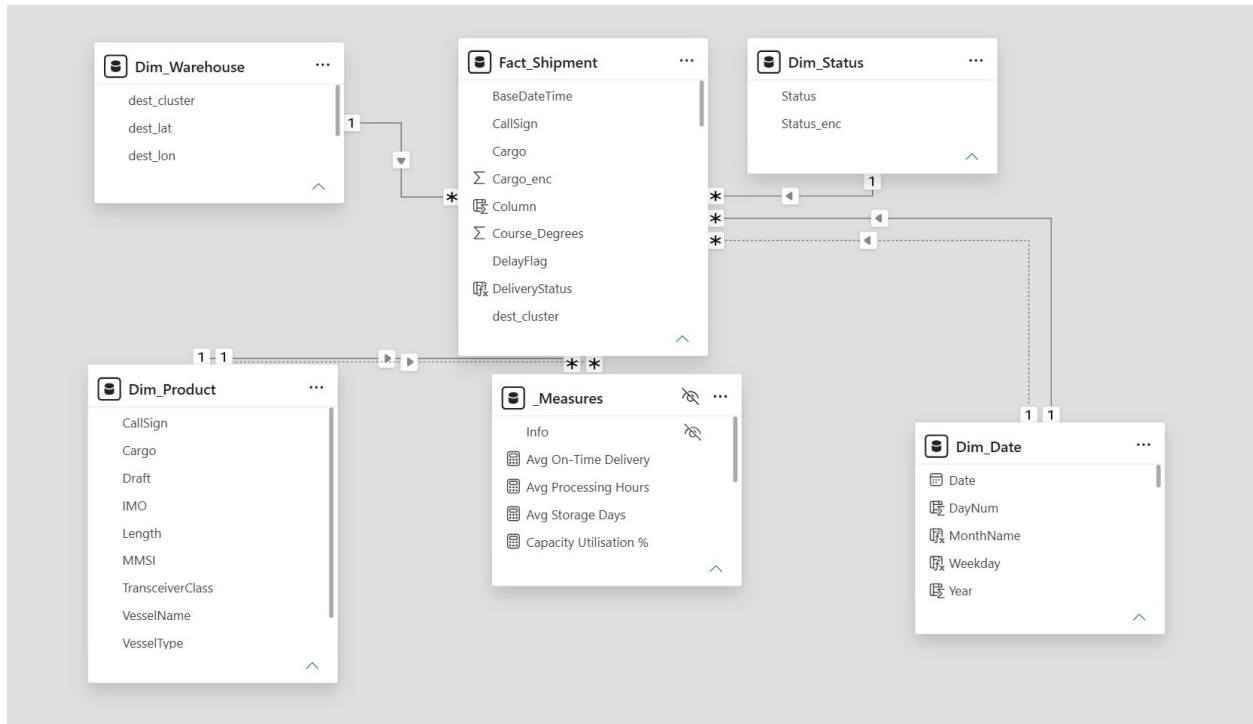


Figure 5: Milestone 4 - Final Star Schema Data Model

7.2 Data Preparation

The available fields were mapped to warehouse-related business concepts using analytical proxies.

Business Metric	Dataset Field / Measure	Purpose
Total Orders	MMSI + dest_cluster	Shipment/order proxy
Shipped Quantity	dist_km	Shipment-flow proxy
Capacity Utilisation	IsMoving	Activity/capacity proxy
On-Time Delivery	DelayFlag	Delivery performance
Picking Accuracy	On-Time Delivery	Accuracy proxy
Processing Hours	ETA_hours	Processing-time proxy
Operating Cost	dist_km × \$2.50	Cost proxy

7.3 DAX Measures

7.3.1 Total Orders

```
Total Orders =  
COUNTROWS(  
    SUMMARIZE(  
        Fact_Shipment,  
        Fact_Shipment[MMSI],  
        Fact_Shipment[dest_cluster]  
    )  
)
```

Purpose : Counts unique combinations of vessel (MMSI) and warehouse (dest_cluster) and uses them as an order/shipment proxy.

7.3.2 Total Shipped Quantity

```
Total Shipped Quantity =  
SUM(Fact_Shipment[dist_km])
```

Purpose: Calculates the total shipment distance using dist_km as a proxy for shipped quantity or shipment flow.

7.3.3 Capacity Utilisation

```
Capacity Utilisation =  
DIVIDE(  
    CALCULATE(  
        COUNTROWS(Fact_Shipment),  
        Fact_Shipment[IsMoving] = "Moving"  
    ),  
    COUNTROWS(Fact_Shipment),  
    0  
)
```

Purpose : Calculates the proportion of shipment records where the vessel is moving, representing operational/capacity utilisation.

7.3.4 Avg On-Time Delivery

```

Avg On-Time Delivery % =
1 -
DIVIDE(
    CALCULATE(
        COUNTROWS(Fact_Shipment),
        Fact_Shipment[DelayFlag] = 1
    ),
    COUNTROWS(Fact_Shipment),
    0
)

```

Purpose: Calculates the percentage of shipment records that were not delayed using DelayFlag.

7.3.5 Picking Accuracy

```

Picking Accuracy =
[Avg On-Time Delivery %]

```

Purpose: Uses the On-Time Delivery percentage as a proxy for picking accuracy because the dataset does not contain actual picking transaction data.

7.3.6 Operating Cost

```

Operating Cost =
SUMX(
    Fact_Shipment,
    Fact_Shipment[dist_km] * 2.5
)

```

Purpose: Estimates operating cost by multiplying shipment distance by the assumed cost rate of \$2.50 per km.

7.3.7 Avg Processing Hours

```

Avg Processing Hours =
AVERAGE(Fact_Shipment[ETA_hours])

```

Purpose: Calculates average ETA and uses it as a proxy for processing or handling time.

7.4 Dashboard 1 – Warehouse Efficiency Report

The dashboard contains six specific KPI cards and seven visualization charts are included in the Warehouse Efficiency Report.

7.4.1 KPI Cards

Six specific KPI cards have been prepared:

- **Total Orders** : This card shows the total number of unique combinations of shipping vessels and warehouses used as the order/shipment component.
- **Total Shipped Quantity** : Total shipping distance used as a proxy for the total quantity shipped is shown.
- **Capacity Utilisation**: This card shows the percentage of the total shipment record moved.
- **On Time Delivery**: This KPI card shows the total percentage of shipment records that are timely.
- **Picking Accuracy**: This card shows the picking accuracy as an indicator based on On-Time Delivery.
- **Operating Cost**: This card shows the estimated operating cost in accordance with the shipping distance.

7.4.2 Total Order Quantity by Warehouse

The data visualized in this graph gives a comparative view of the order and shipment activity for the various warehouse locations. This allows for the identification of the share of shipment activity at different warehouses.

7.4.3 Capacity Utilisation by Warehouse

The plotting shown in this graph helps in giving a comparative view of the capacity utilization proxy of the warehouse location. This was calculated based on the percentage of records in which the vessel could be classified as being in Moving state.

7.4.4 Picking Accuracy by Warehouse

The representation given in the graph compares picking accuracy proxy in the various warehousing locations. The On-Time Delivery measure was taken as a proxy for picking accuracy since picking transaction data was unavailable.

7.4.5 Total Orders by Zone

A donut chart displays the distribution of orders across operational zones:

- Dispatch
- Packing
- Picking
- Receiving
- Storage

This provides a visual breakdown of order activity across the different zones.

7.4.6 Average On-Time Delivery by Warehouse

This chart compares On-Time Delivery percentages across warehouse locations.

It shows the proportion of shipment records that were not delayed for each warehouse.

7.4.7 Average Processing Hours by Warehouse

This chart compares the average processing hours across warehouses.

The value is calculated using the average ETA_hours as a processing-time proxy.

7.4.8 Total Product Category

A donut chart displays the distribution of product/cargo categories.

The categories include:

- Fashion
- Grocery
- Electronics
- Industrial
- Home Appliances

This visual provides a category-level view of the shipment data



Figure 6: Milestone 4 — Warehouse Efficiency Report Dashboard (Power BI)

7.5 Dashboard 2 – Executive Overview

The Executive Overview dashboard features a total of six KPI cards, five visual graph patterns, and four different slicers.

The dashboard contains:

- Total Orders
- Total Shipped Quantity
- Capacity Utilization
- On-Time Delivery
- Picking Accuracy
- Operating Cost

These provide a quick look at the selected data.

7.5.1 Date Slicer

The Date slicer helps users choose a specific date or a range of dates.

The executive overview dashboard visualizes the selections in real-time.

7.5.2 Warehouse Slicer

The Warehouse slicer helps users select specific warehouses for studying purposes.

7.5.3 Zone Slicer

The Zone slicer helps users filter the dashboard in terms of:

- Dispatch
- Packing
- Picking
- Receiving
- Storage

7.5.4 Order Type Slicer

The Order Type slicer contains:

- Bulk
- Express
- Standard

It helps users analyze the dashboard for various order types.

7.5.5 Total Orders by Date

This chart shows the order or shipment activity over time.

It lets users track order activity changes over dates.

7.5.6 Total Orders by Warehouse

This chart compares order or shipment totals across warehouses.

7.5.7 Total Inventory by Warehouse

This graph visualizes inventory measure across facility locations, showing how inventory flows between them.

7.5.8 On-Time Delivery by Warehouse

This graph shows delivery efficiency across warehouse locations, based on the On-Time Delivery measure.

7.5.9 Picking Accuracy by Warehouse

This graph compares picking accuracy measures across various warehouse locations.

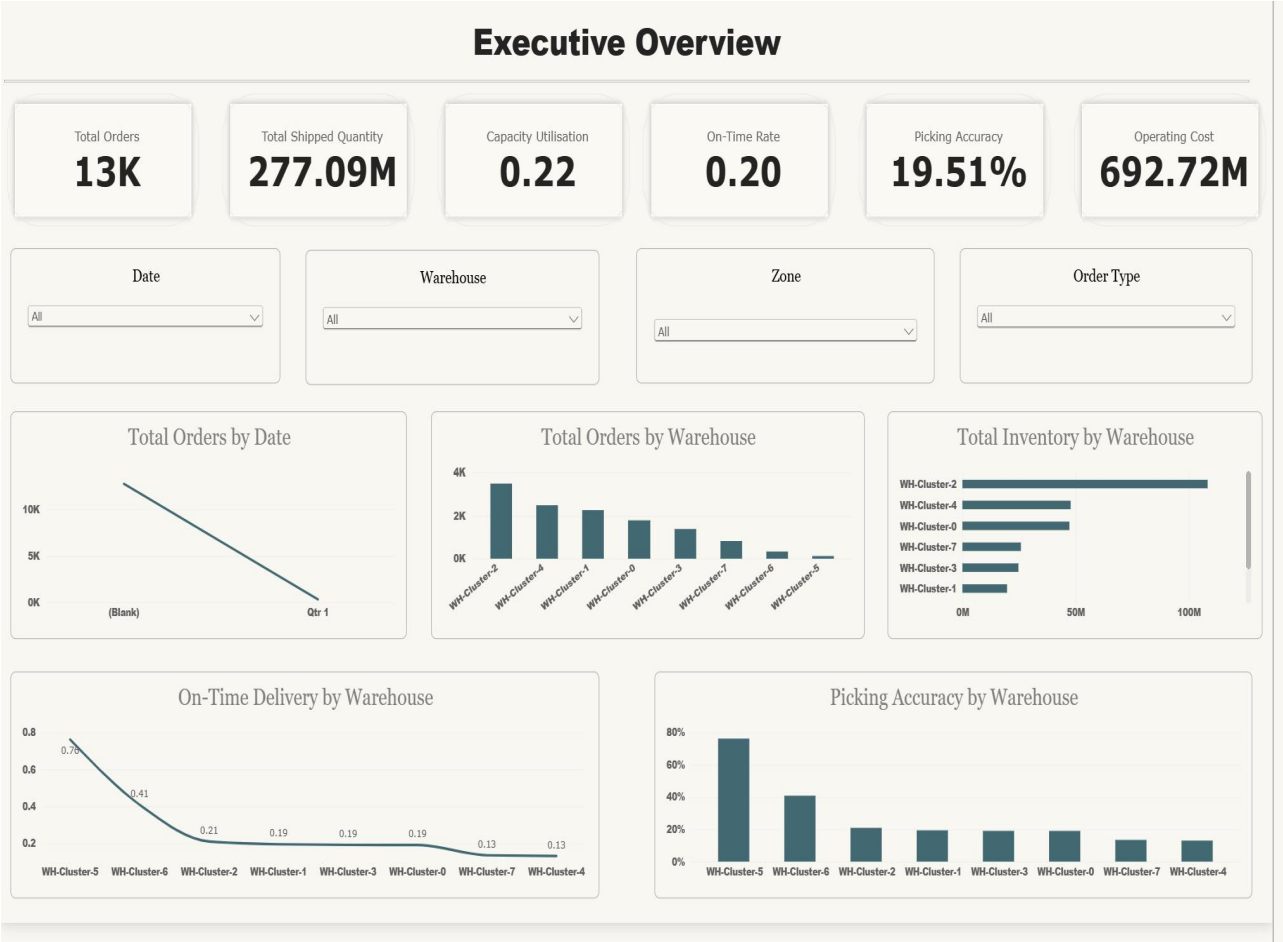


Figure 7: Milestone 4 — Executive Overview Dashboard (Power BI)

7.6 Dashboard 3 – Performance Optimization

The Performance Optimization dashboard shows the performance of the warehouse through the use of graphs.

7.6.1 Warehouse City

The dashboard shows the warehouse location in a city manner.

7.6.2 Capacity Utilisation

The dashboard shows the capacity utilization index of the warehouse in operation.

7.6.3 On-time Delivery

The dashboard shows how many shipment records are not delayed.

7.6.4 Picking Accuracy

The dashboard shows the picking accuracy rate of the warehouse.

7.6.5 Processing Hours

The dashboard shows average processing time according to ETA.

The table combines all five performance indicators to increase owners awareness of the warehouse location.

Dashboard Performance Optimization				
Warehouse City	Capacity Utilisation	On-Time Delivery	Picking Accuracy	Avg Processing Hours
WH-Seattle	0.17	0.19	19.41%	55.58
WH-Santa Barbara	0.20	0.19	19.07%	61.54
WH-San Juan	0.33	0.41	40.87%	45.17
WH-New York	0.24	0.19	19.03%	60.54
WH-New Orleans	0.24	0.21	20.93%	60.30
WH-Miami	0.16	0.13	13.05%	64.39
WH-Honolulu	0.30	0.76	76.24%	28.05
WH-Chicago	0.27	0.13	13.46%	64.06
Total	0.22	0.20	19.51%	60.07

Figure 8: Milestone 4 — Performance Optimization Dashboard (Power BI)

Conclusion

The project on Supply Chain Visibility System with Optimization Analytics was accomplished in four organized milestones through Power BI and a shipment dataset from the AIS platform.

Milestone 1 created the project base with the help of vessel movement and meteorological data, the data model development, as well as primary KPIs in the area of supply chain, including the number of vessels, shipment activities, average speed of movement, distance covered, and impact of weather.

Milestone 2 moved the focus of analysis to that area of inventory, product, warehouse, and delivery management, and performance indicators were created in relation to inventory, throughput data, shipment activities, delays in the process of delivery, and fulfillment.

Milestone 3 of the project concentrated on the performance of transport and suppliers/carriers. The analysis involved information such as transportation cost, average shipment cost, on-time rate, fulfillment rate, and quality measurements in Power BI.

Milestone 4 concluded the project by enhancing the existing model by including warehouse efficiency, executive reporting, and process improvement. Three dashboards were developed: Warehouse Efficiency Report, Executive Overview, and Performance Improvement. DAX measures were adopted for order activity, shipment flow, capacity utilisation, delivery quality, processing time, and estimated operation cost.

The project shows how one can prepare data, create dimensional models, apply DAX measures, visualize data, and design dashboards to enhance the visibility of supply chain processes. The execution of the project demonstrates another practical use of the proxy measures when it is impossible to get some database fields from the original AIS database.

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